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jtleek / genstats

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Branch: gh-pages genstats / inst / doc / 02_06_galton-example.R

jtleek 12 days ago big update

1 contributor

86 lines (69 sloc) 3.2 KB

RawBlameHistory

```
1  ## ----global_palette, results = 'asis'-----
2  rm(list=ls())
3  tropical= c('darkorange', 'dodgerblue', 'hotpink', 'limegreen', 'yellow')
4  palette(tropical)
5
6  ## ----global_options,warning=FALSE,message=FALSE-----
7  ## see ch. 10 Hooks of Xie's knitr book
8  library(knitr)
9  knit_hooks$set(setPch = function(before, options, envir) {
10    if(before) par(pch = 19)
11  })
12  opts_chunk$set(setPch = TRUE)
13
14  ## ----global_plot,warning=FALSE, message=FALSE-----
15  knitr::opts_chunk$set(fig.width=5, fig.height=5, size="footnotesize",
16                        warning=FALSE, message=FALSE)
17  knitr::knit_hooks$set(small.mar = function(before, options, envir) {
18    if (before) graphics::par(mar = c(5,5,1.5,1))
19  })
20
21  ## ----load_hidden, echo=FALSE, results="hide", warning=FALSE-----
22  suppressPackageStartupMessages({
23    library(devtools)
24    library(Biobase)
25    library(UsingR)
26  })
27
28  ## ----load-----
29    library(devtools)
30    library(Biobase)
31    library(UsingR)
32
33  ## ----install_packages, eval=FALSE-----
34  # install.packages(c("devtools", "UsingR"))
35  # source("http://www.bioconductor.org/biocLite.R")
36  # biocLite(c("Biobase"))
37
38  ## ----galton,fig.height=4,fig.width=8-----
39  data(galton)
40  par(mfrow=c(1,2))
41  hist(galton$child,col="blue",breaks=100)
42  hist(galton$parent,col="blue",breaks=100)
43
44  ## ---- dependson="galton",fig.height=4,fig.width=4-----
45  hist(galton$child,col="blue",breaks=100)
46
47  ## ---- dependson="galton",fig.height=4,fig.width=4-----
48  hist(galton$child,col="blue",breaks=100)
49  meanChild <- mean(galton$child)
50  lines(rep(meanChild,100),seq(0,150,length=100),col="red",lwd=5)
51
52  ## ---- dependson="galton",fig.height=4,fig.width=4-----
53  plot(galton$parent,galton$child,pch=19,col="blue")
54
55  ## ---- dependson="galton",fig.height=4,fig.width=4-----
56  plot(galton$parent,galton$child,pch=19,col="blue")
57  near65 <- galton[abs(galton$parent - 65)<1, ]
58  points(near65$parent,near65$child,pch=19,col="red")
59  lines(seq(64,66,length=100),rep(mean(near65$child),100),col="red",lwd=4)
60
61  ## ---- dependson="galton",fig.height=4,fig.width=4-----
62  plot(galton$parent,galton$child,pch=19,col="blue")
63  near71 <- galton[abs(galton$parent - 71)<1, ]
```

```
64 points(near71$parent,near71$child,pch=19,col="red")
65 lines(seq(70,72,length=100),rep(mean(near71$child),100),col="red",lwd=4)
66
67 ## ---- dependson="lm1",fig.height=4,fig.width=4-----
68 plot(galton$parent,galton$child,pch=19,col="blue")
69 lm1 <- lm(galton$child ~ galton$parent)
70 lines(galton$parent,lm1$fitted,col="red",lwd=3)
71
72 ## ---- dependson="lm1",fig.height=4,fig.width=4-----
73 plot(galton$parent,galton$child,pch=19,col="blue")
74 lines(galton$parent,lm1$fitted,col="red",lwd=3)
75
76 ## ---- dependson="lm1",fig.height=4,fig.width=8-----
77 par(mfrow=c(1,2))
78 plot(galton$parent,galton$child,pch=19,col="blue")
79 lines(galton$parent,lm1$fitted,col="red",lwd=3)
80 plot(galton$parent,lm1$residuals,col="blue",pch=19)
81 abline(c(0,0),col="red",lwd=3)
82
83 ## ----session_info-----
84 devtools::session_info()
85
```

